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Hydrodynamic regulation drives azo dye transformation and hydrolysis acidification by reshaping microbial metabolism in the micro-aerobic hydrolysis acidification reactor

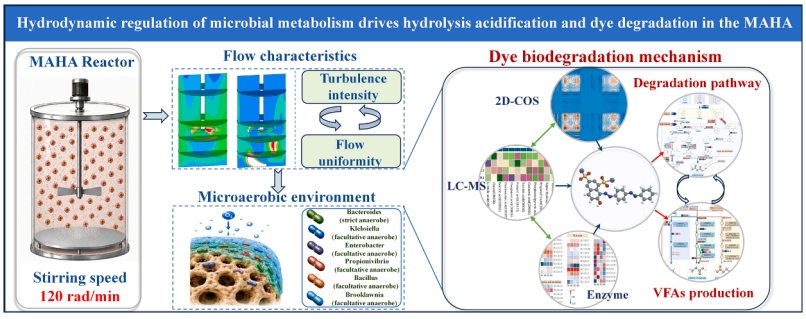
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HIGHLIGHTS

- R2 (120 rad/min) balanced turbulence intensity and flow uniformity.
- Flow enhanced mass transfer and enriched facultative fermentative bacteria.
- The degradation pathway of Acid Red GR was elucidated.
- R2 (120 rad/min) achieved the highest VFA production and biodegradability.
- The flow driven metabolic mechanism in the MAHA reactor was revealed.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Acid Red GR is a representative sulfonated azo dye frequently detected in dyeing wastewater, yet the role of hydrodynamic conditions in regulating its biodegradation in micro-aerobic hydrolysis acidification (MAHA) systems remains unclear. This study investigated how hydrodynamic conditions drive Acid Red GR transformation, hydrolysis acidification, and microbial metabolic responses in the MAHA reactor. The results showed that a moderate stirring speed of 120 rad/min (R2) balanced turbulence intensity and flow uniformity, with relatively high oxygen transfer, a favorable micro-aerobic niche, and the highest biodegradability (0.50) and volatile fatty acid (VFA) production (140.38 mg/L). This hydrodynamic condition enriched hydrolytic, facultative, and fermentative genera, including *Klebsiella*, *Enterobacter*, *Bacteroides*, *Propionivibrio*, and *Bacillus*, and was associated with higher abundances of genes encoding enzymes involved in azo reduction, aromatic intermediate transformation, pyruvate generation, and acidogenic metabolism. Integrated metagenomic, metabolomic, and functional group evidence further indicated that Acid Red GR degradation proceeded through azo reduction, deamination, desulfonation, aromatic ring cleavage, and subsequent acidogenesis. These findings

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demonstrate that hydrodynamic optimization regulates MAHA performance by coordinating oxygen transfer, microbial niche formation, and metabolic functional potential, providing a mechanistic basis for enhancing the biodegradability and treatment efficiency of dyeing wastewater.

1. Introduction

Acid Red GR is a representative sulfonated azo dye contaminant frequently detected in dyeing wastewater [1]. Its highly conjugated azo structure and sulfonated aromatic framework contribute to persistent coloration, limited biodegradability, and resistance to conventional biological treatment [2]. Moreover, incomplete transformation of azo dyes may lead to the accumulation of aromatic amines and other intermediates with potential ecological risks [3]. Therefore, developing effective strategies that not only remove dye color but also promote complete molecular transformation and detoxification of azo dye contaminants remains an important challenge in wastewater treatment.

Micro-aerobic hydrolysis acidification (MAHA) has recently emerged as a promising biological approach for enhancing azo dye transformation and improving wastewater biodegradability [1,4]. Unlike conventional anaerobic hydrolysis acidification, MAHA introduces trace oxygen to maintain a dissolved oxygen (DO) concentration of 0.1–1.0 mg/L, creating a micro-aerobic environment that may simultaneously facilitate anaerobic azo-bond reduction, aerobic oxidation of aromatic intermediates, and microbial carbon metabolism [5,6]. Previous studies have shown that MAHA can achieve over 80% azo dye decolorization, substantial organic matter removal (30–55%), and enhanced biodegradability ($BOD_5/COD = 0.36–0.50$), while promoting volatile fatty acid (VFA) production and the further transformation of refractory aromatic intermediates through processes such as aromatic-ring cleavage [1,7,8]. However, within this narrow oxygen range, even subtle variations in the local micro-aerobic environment may substantially affect redox conditions and biological responses [9]. Thus, understanding microenvironmental regulation is critical for optimizing MAHA performance.

Hydrodynamic conditions play a critical role in shaping the micro-aerobic environments characteristic of MAHA systems [10,11]. By regulating gas-liquid mass transfer, bubble dispersion, and oxygen transport, hydrodynamics can reshape the local microenvironments experienced by functional microorganisms and consequently affect their metabolic activity and pollutant transformation [12–14]. Existing studies on hydrolysis acidification reactors have shown that hydrodynamic optimization can improve mixing, mitigate dead zones and short-circuiting, and enhance treatment performance [15,16]. However, the specific role of hydrodynamics in regulating micro-aerobic environments in MAHA systems remains insufficiently resolved, and hydrodynamic investigation has therefore been proposed as an important direction for future research [17]. This issue is particularly important because recent MAHA studies have demonstrated that changes in micro-aerobic conditions can substantially alter microbial communities, functional genes, metabolic pathways, and the transformation of refractory organic matter [1,4]. For azo dyes, Acid Red GR has been proposed to undergo azo-bond reduction, desulfonation, hydrolytic deamination, aromatic-ring cleavage, and subsequent conversion through central carbon metabolism [1]. However, different micro-aerobic conditions have been reported to result in two distinct pollutant transformation pathways, dominated by hydroxylation and deamination, respectively [9]. Corresponding changes were also observed in microbial functions, including genes and enzymes involved in phenolic oxidation and aromatic-ring cleavage. Therefore, elucidating microbial metabolism and azo dye degradation mechanisms in MAHA systems remains a important issue that requires further investigation.

Resolving these mechanisms requires an integrated approach that links hydrodynamic regulation with microbial metabolic responses and molecular transformation of azo dyes. Metagenomic analysis can characterize microbial functional potential, whereas liquid chromatography-mass spectrometry (LC-MS)-based metabolite identification can reveal transformation intermediates [18,19]. However, these approaches do not directly resolve the relative sequence in which functional groups respond to hydrodynamic perturbation. Two-dimensional correlation spectroscopy (2D-COS), with turbulence intensity used as a continuous external perturbation, can complement these analyses by enhancing the resolution of overlapping Fourier transform infrared spectroscopy (FTIR) bands and determining the relative response sequence of functional groups [4,20]. Although hydrodynamic analysis, metagenomic analysis, LC-MS-based metabolite identification, and 2D-COS have each been applied in related environmental and bioprocess studies [18,21], their integration for resolving hydrodynamic-microbial-molecular interactions in MAHA systems remains limited. Such integration is needed to connect reactor-scale flow characteristics and oxygen transfer with microbial functional responses, metabolite profiles, and the structural transformation of dye-derived organics.

This study examined how hydrodynamic conditions regulate the biodegradation pathway and metabolic transformation of Acid Red GR in the MAHA reactor treating dyeing wastewater. Reactor-scale flow characteristics were resolved using the Euler-Euler two-fluid model, while the corresponding oxygen transfer responses were quantified experimentally. Process performance and EPS characteristics were further evaluated to characterize dye transformation, biodegradability improvement, and acidogenic behavior under different hydrodynamic conditions. Metagenomic analysis and LC-MS-based metabolite identification were integrated with 2D-COS to relate microbial functional potential and transformation intermediates to the response sequence of functional groups. These complementary datasets were further used to elucidate Acid Red GR degradation and acidogenic pathways and to clarify the links among flow regulation, oxygen transfer, microbial metabolism, and dye-derived organic matter transformation in the MAHA system. These findings provide a mechanistic basis for hydrodynamic regulation of MAHA pretreatment by linking flow and oxygen transfer with microbial metabolism and dye-derived molecular transformation.

2. Materials and methods

2.1. Setup and operation of the MAHA reactor

Four identical plexiglass MAHA reactors were operated in parallel at ambient laboratory temperature. Each reactor had an internal diameter of 23.4 cm, a height of 70.0 cm and an effective volume of 25 L (Fig. S1). The reactors were equipped with stirring, aeration, influent and effluent systems, and an online DO monitoring system. Air was delivered to each reactor via an aeration disc after passing through a flowmeter and gas chamber. The aeration holes had a diameter of 0.16 mm and were spaced 10 mm apart. Polyurethane carriers ($1 \times 1 \times 1$ cm) were used as biofilm supports, with a filling ratio of 30% based on the effective reactor volume.

Seed sludge was collected from the secondary clarifier of the Xi'an Fifth Wastewater Treatment Plant. Before inoculation, the sludge was acclimated for 3–5 d with synthetic domestic wastewater at a C:N:P ratio of 100:5:1, and the sludge concentration was adjusted to 3000–4000 mg/L before introduction into the four reactors. During start-up, DO was controlled at 0.1–0.3 mg/L until treatment performance stabilized and biofilms were established.

The synthetic dyeing wastewater was prepared based on a survey of

water quality at a dyeing wastewater treatment plant in Zhejiang Province, China. Acid Red GR and polyvinyl alcohol (PVA) were added at 30 and 300 mg/L, respectively, while polyethylene glycol 4000, sodium dodecyl sulfate, sodium dodecyl sulfonate, and glucose were each added at 100 mg/L. Nitrogen, phosphorus, trace elements, and other nutrients were dosed according to a previously reported method [1]. To establish different hydrodynamic conditions, stirring speeds were set at 100, 120, 140, and 160 rad/min for R1, R2, R3, and R4, respectively, with 100 rad/min identified experimentally as the critical stirring speed for carrier fluidization. During operation, the hydraulic retention time (HRT) was 8 h, with an aeration rate of 0.6 $m^3/(h{\cdot}m^3)$. The DO, oxidation-reduction potential (ORP), and pH were monitored, while influent and effluent chemical oxygen demand (COD), volatile fatty acids (VFAs), PVA, color, and five-day biochemical oxygen demand (BOD_5) were measured.

2.2. Flow characteristic simulation methods

Computational fluid dynamics (CFD) simulations were performed in ANSYS Fluent software to resolve the hydrodynamic characteristics of the MAHA reactor. A 1:1 three-dimensional model (Fig. S2a) was constructed based on the experimental MAHA reactor and simplified for numerical simulation. The geometry was meshed in ICEM software using unstructured grids, and grid independence analysis showed that 221,318 cells were sufficient for reliable simulation (Fig. S2b).

The reactor contained gas, liquid, and suspended biofilm carriers. To reduce model complexity, the dyeing wastewater and biofilm carriers were treated as an equivalent continuous phase, because the carriers were mainly transported with the bulk liquid flow and the present work focused on gas dispersion, liquid mixing, and oxygen transfer rather than carrier-scale collision or trajectory dynamics. Therefore, the simulated system was simplified as a gas-liquid two-phase flow. The Euler-Euler two-fluid model was used to describe the gas and equivalent liquid phases, and the mass and momentum conservation equations were solved for each phase to characterize phase distribution, flow transport, and interphase momentum exchange (Table S1). The realizable k-ε turbulence model was applied to resolve turbulent flow characteristics. A population balance model (PBM) was further coupled with the CFD framework to account for bubble coalescence and breakup, which determine bubble size distribution and gas-liquid interfacial area (Table S1). The CFD-PBM coupled model was benchmarked against experimental hydrodynamic data from a similar reactor [22], and the model predictions showed reasonable agreement with the reported measurements (Fig. S2c). Because the experimental data were obtained from a similar rather than the present reactor, this comparison should be regarded as indirect rather than reactor-specific validation.

2.3. Oxygen mass transfer measurement

DO in the reactor was continuously monitored at 1 -min intervals using a four-channel optical fiber measuring instrument (OXY-4 SMA trace, PreSens GmbH, Regensburg, Germany). Anhydrous sodium sulfite was first added to deplete DO, after which aeration was initiated and the DO recovery to saturation was recorded. The volumetric oxygen mass transfer coefficient (k_La) was calculated from the DO recovery curve according to Liu et al. (2026) [23]. The standard oxygen mass transfer rate (SOTR) and standard oxygen mass transfer efficiency (SOTE) were further calculated using the saturated DO concentration under standard conditions and the aeration rate [24]. Under stable operation, the oxygen mass transfer rate (OTR) was assumed to be equal to the microbial oxygen uptake rate (OUR) [25]. After aeration was stopped, the decline in DO was mainly attributed to microbial oxygen consumption. Therefore, OUR was calculated from the slope of the DO-time curve during the non-aeration phase and was used as the steady-state OTR.

2.4. Water quality analysis

ORP and pH were measured using a portable Hach meter (Hach, USA). COD and BOD_5 were determined using the potassium dichromate method and the five-day incubation method, respectively, as previously reported [4]. Dye concentration and color variation were characterized by measuring absorbance at 509 nm using a UV–visible spectrophotometer. PVA concentration was determined by UV spectrophotometry at 680 nm and calculated from a standard curve [1]. VFAs were analyzed using a PerkinElmer Altus high-performance liquid chromatography (HPLC) system equipped with an Aminex HPX-87H column (300 × 7.8 mm; Bio-Rad, USA). Before injection, samples were filtered through a 0.22 μm membrane and degassed for 15 min. A 0.005 mol/L H_2SO_4 solution was used as the mobile phase at a flow rate of 0.6 mL/min. Individual VFA species were identified based on their retention times and quantified according to the corresponding peak areas using external calibration curves.

The fluorescence characteristics of influent and effluent dissolved organic matter (DOM) were measured using a high-sensitivity integrated fluorescence spectrometer (FluoroMax-4, HORIBA Scientific, USA). The excitation and emission wavelength ranges were 240–550 nm and 290–600 nm, respectively, with a step size of 5 nm and a slit width of 3 nm. Based on excitation-emission matrix (EEM) data, fluorescence regional integration was used to quantitatively analyze different fluorescent DOM components, according to previous research [26].

2.5. EPS extraction and characterization

Biofilm was collected from the carriers by manual squeezing. EPS fractions, including soluble EPS (SL-EPS), loosely bound EPS (LB-EPS) and tightly bound EPS (TB-EPS), were extracted from the biofilm using a thermal extraction method [27]. Proteins (PN) and polysaccharides (PS) in EPS were determined using the modified Folin-Lowry method and the anthrone method, respectively [28].

2.6. Functional group characterization

EPS samples from different layers were scanned using a Fourier transform infrared spectrometer (VERTEX 70, Bruker, Germany) over the range of 4000–400 cm^{-1}. Functional groups were qualitatively identified according to peak positions. 2D-COS was then used to analyze the response characteristics of these functional groups under flow perturbation. The raw spectra were processed by baseline correction, smoothing and normalization. Turbulence intensity was used as the external continuous perturbation variable to construct synchronous and asynchronous two-dimensional correlation spectra. The response sequence of characteristic functional groups was determined from the sign relationship between synchronous and asynchronous peaks according to Noda's rules [29].

2.7. Metagenomic analysis

At the end of each operating condition, carriers were collected from several positions within the reactor. Biofilms detached from these carriers were combined and thoroughly homogenized to obtain one composite sample for each condition. Accordingly, four composite biofilm samples corresponding to R1-R4 were subjected to metagenomic analysis. The samples were stored at −80 °C before DNA extraction. Total genomic DNA was extracted using a commercial biofilm DNA extraction kit, and DNA concentration, purity, and integrity were evaluated prior to library preparation. Qualified DNA was fragmented for paired-end library construction and subsequently subjected to high-throughput sequencing on an Illumina platform (Personalbio, Shanghai, China). Raw sequencing reads were filtered to remove adapter sequences, low-quality reads, and short sequences, followed by de novo assembly using MEGAHIT. Coding sequences were predicted using Prodigal, and

the predicted genes were clustered to generate a nonredundant gene catalog. Gene abundances were estimated by mapping the quality-filtered reads to the gene catalog. Functional annotation was performed against the KEGG database to obtain KEGG Orthology (KO) and pathway information. Genes annotated as encoding enzymes potentially involved in Acid Red GR transformation and acidogenesis were subsequently screened, and their relative abundances and taxonomic affiliations were used to evaluate the functional potential of different microbial taxa.

2.8. Metabolite detection and identification

Samples for metabolite analysis were collected from each reactor (R1-R4) at the end of operation and analyzed by LC-MS. After thawing at 4 °C, samples were extracted with methanol and centrifuged at 12,000 rpm for 10 min at 4 °C. The supernatants were dried, reconstituted in 150 μL of 2-chloro-L-phenylalanine solution (4 ppm in 80% methanol–water), and filtered through a 0.22 μm membrane.

Metabolites were analyzed using an Agilent 1290 Infinity LC system coupled with an AB SCIEX TripleTOF 6600 mass spectrometer.

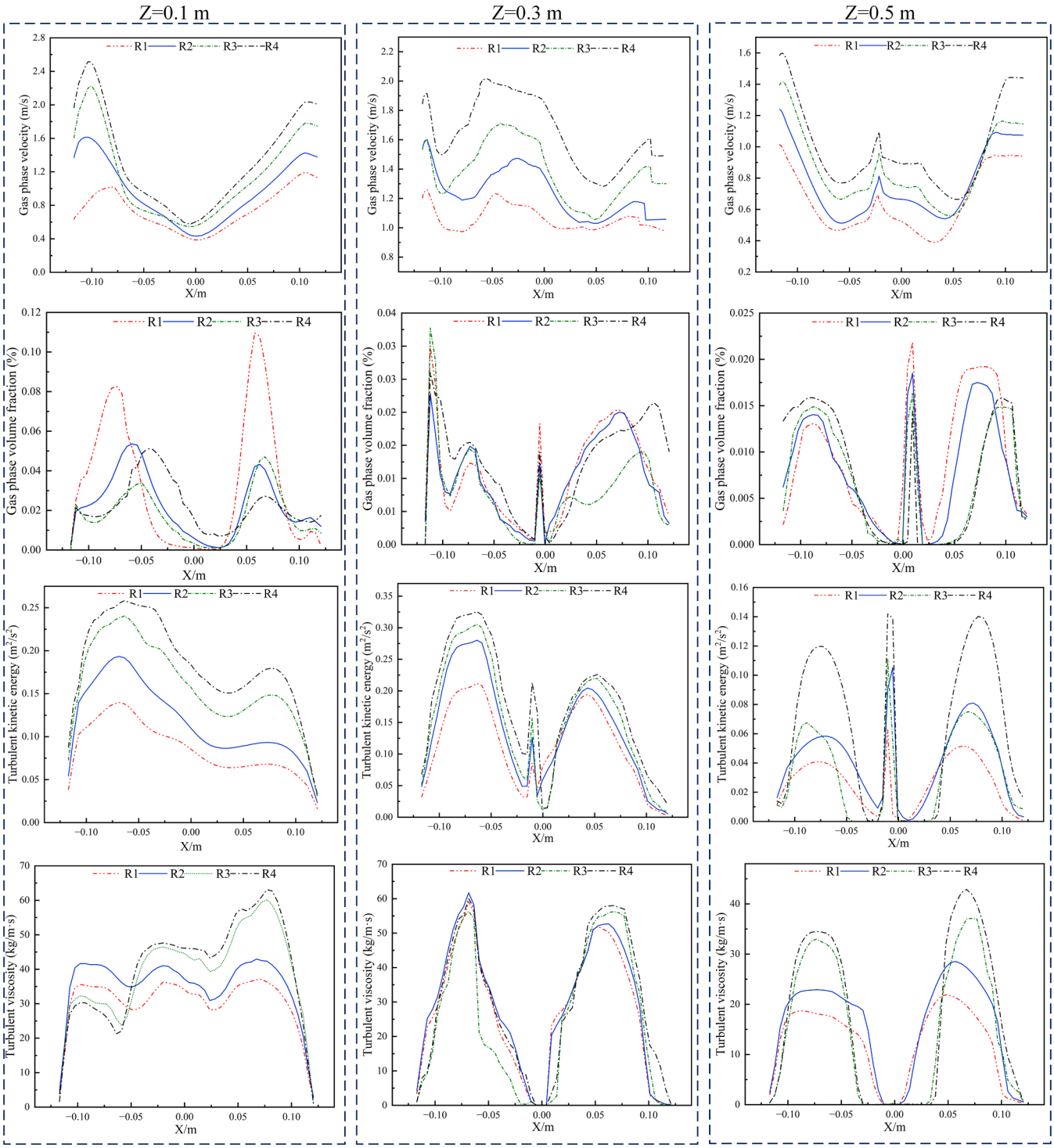


Fig. 1. Gas-phase velocity, gas-phase volume fraction, turbulent kinetic energy, and turbulent viscosity at Z = 0.1, 0.3, and 0.5 m.

Separation was performed on an ACQUITY UPLC BEH Amide column (1.7 μm, 2.1 × 100 mm) at 25 °C using 25 mM ammonium acetate and 25 mM ammonium hydroxide in water (A) and acetonitrile (B) at 0.5 mL/min. A 12-min gradient from 95% to 40% B was applied, followed by re-equilibration. Mass spectra were acquired in positive and negative electrospray ionization modes, with TOF-MS and MS/MS scan ranges of *m/z* 60–1000 and 25–1000, respectively. Raw data were converted using ProteoWizard and processed with XCMS. Metabolites were identified based on accurate mass (error < 25 ppm), retention time, MS/MS fragmentation, and collision-energy information against an in-house database, followed by manual verification.

2.9. Statistical Analysis

Correlation networks among functional bacteria, key enzymes, and metabolites were constructed in Python using pandas, scipy.stats, networkx, and matplotlib. Mantel tests were performed in R to assess relationships among environmental factors, microbial communities, enzymes, and metabolites. Pearson correlations with $|r| > 0.5$ were considered strong. Group differences were evaluated using one-way ANOVA with Tukey's HSD test or the Kruskal-Wallis test with Dunn-Holm correction, as appropriate. Statistical significance was set at $p < 0.05$, and different lowercase letters indicated significant differences among groups.

3. Results and discussion

3.1. Hydrodynamic characteristics under different stirring conditions

Increasing stirring speed progressively altered the gas-phase flow field, as reflected by increases in gas-phase velocity, turbulent kinetic energy, and turbulent viscosity (Fig. 1). These changes indicated enhanced gas-liquid mixing and momentum exchange under stronger stirring. However, the higher stirring conditions (R3 and R4) generated more pronounced local peaks in velocity and turbulence-related parameters, suggesting increased spatial heterogeneity of the flow field. Such localized high-velocity and high-turbulence regions may promote rapid bubble transport and increase local shear, potentially altering the spatial persistence and distribution of oxygen within the reactor. In contrast, R2 maintained sufficient flow intensity while exhibiting a more continuous distribution of turbulent viscosity, indicating a more spatially uniform hydrodynamic field. Compared with the higher stirring conditions, R2 therefore provided a relatively favorable combination of flow intensity and spatial uniformity. Because direct hydrodynamic measurements were not available for validation in the present reactor, the CFD results were interpreted primarily in terms of relative differences among R1-R4 rather than as exact predictions of local hydrodynamic quantities. The experimentally determined oxygen-transfer responses were subsequently used as independent evidence to evaluate whether the hydrodynamic differences predicted by CFD were associated with changes in gas-liquid oxygen transfer.

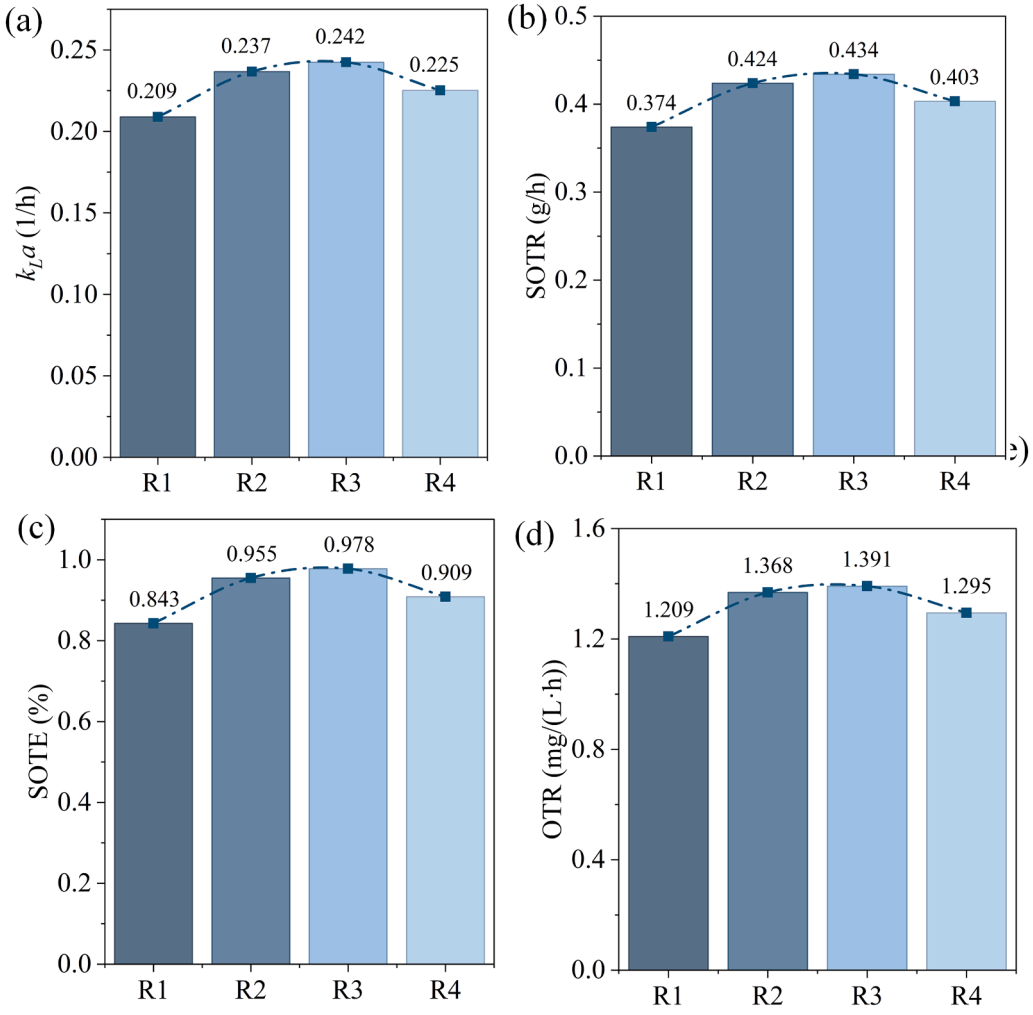


Fig. 2. Oxygen mass transfer characteristics of the reactors.

3.2. Oxygen mass transfer characteristics

Oxygen mass transfer showed a nonlinear response to increasing turbulence intensity (Figs. 2a-2c). The k_La, SOTR, and SOTE increased from 0.21 h^{-1}, 0.38 g/h, and 0.84% in R1 to 0.24 h^{-1}, 0.42 g/h, and 0.95% in R2, respectively. These parameters changed only slightly from R2 to R3 and subsequently decreased to 0.23 h^{-1}, 0.40 g/h and 0.91% in R4. OTR showed a similar trend, increasing from 1.21 mg/(L·h) in R1 to 1.37 and 1.39 mg/(L·h) in R2 and R3, respectively, before decreasing to 1.29 mg/(L·h) in R4. This trend was consistent with previous research, in which oxygen transfer performance was significantly affected by turbulence intensity and an effective stirring speed range was observed [30]. In the present system, oxygen transfer was enhanced from R1 to R2, whereas further increases in stirring intensity produced little additional improvement in R3. The decline observed in R4 may be associated with the stronger and more heterogeneous flow field identified in Section 3.1, which could accelerate bubble transport and gas escape and shorten effective gas-liquid contact time [31].

The DO concentration increased progressively across R1-R4 (Fig. S3), reaching 0.11 ± 0.01, 0.14 ± 0.02, 0.18 ± 0.02, and 0.24 ± 0.04 mg/L, respectively. This was close to the 0.18 ± 0.05 mg/L reported for a micro-aerated upflow anaerobic sludge blanket (UASB) reactor treating synthetic textile wastewater [32], indicating that the present system operated within a comparable low-DO micro-aerobic regime. Despite the progressive increase in bulk DO from R2 to R4, oxygen transfer showed only limited further improvement, indicating that bulk DO alone did not determine oxygen transfer performance. R2 achieved oxygen transfer comparable to R3 at a lower DO level while maintaining a more uniform flow field, suggesting that moderate stirring provided a favorable balance between oxygen delivery and flow uniformity.

3.3. Pollutant transformation and acidogenesis

The pollutant transformation characteristics under different stirring conditions are shown in Fig. 3. Decolorization efficiencies remained consistently high across R1-R4, ranging from 87.13% to 89.38%, with no significant differences among the reactors ($p = 0.122$; Fig. 3a). The consistently high decolorization across R1-R4 suggested that the low-oxygen conditions were sufficient to sustain effective azo-dye transformation, which was further supported by the azo reduction-related functional potential identified in Section 3.8. In parallel, 3D-EEM fluorescence spectroscopy revealed substantial transformation of DOM during MAHA treatment (Fig. S4). The influent exhibited distinct fluorescence signals in regions II, III, IV, and V, corresponding to aromatic protein-like, fulvic acid-like, soluble microbial byproduct-like, and humic acid-like substances, respectively. These fluorescence signals were markedly attenuated after treatment, indicating substantial transformation of fluorescent organic constituents. In contrast to the high and relatively stable decolorization, COD and PVA removal efficiencies remained comparatively low across all reactors, ranging from 18.86% to 22.19% and 16.68–18.61%, respectively (Fig. 3b-c), consistent with values reported for other MAHA systems [1]. This apparent discrepancy between efficient decolorization and limited COD removal suggests that MAHA primarily promoted structural transformation and partial conversion of refractory organics rather than complete mineralization. In particular, azo-dye cleavage and subsequent hydrolysis/acidification can convert complex organic structures into smaller and more biodegradable intermediates [7]. Accordingly, wastewater biodegradability increased markedly after MAHA treatment (Fig. 3d). Notably, the BOD_5/COD ratio in R2 reached 0.50, exceeding the value reported in a previous MAHA study [4], where BOD_5/COD increased from 0.32 to 0.36. Collectively, these results indicate that the MAHA process effectively transformed refractory organic constituents into

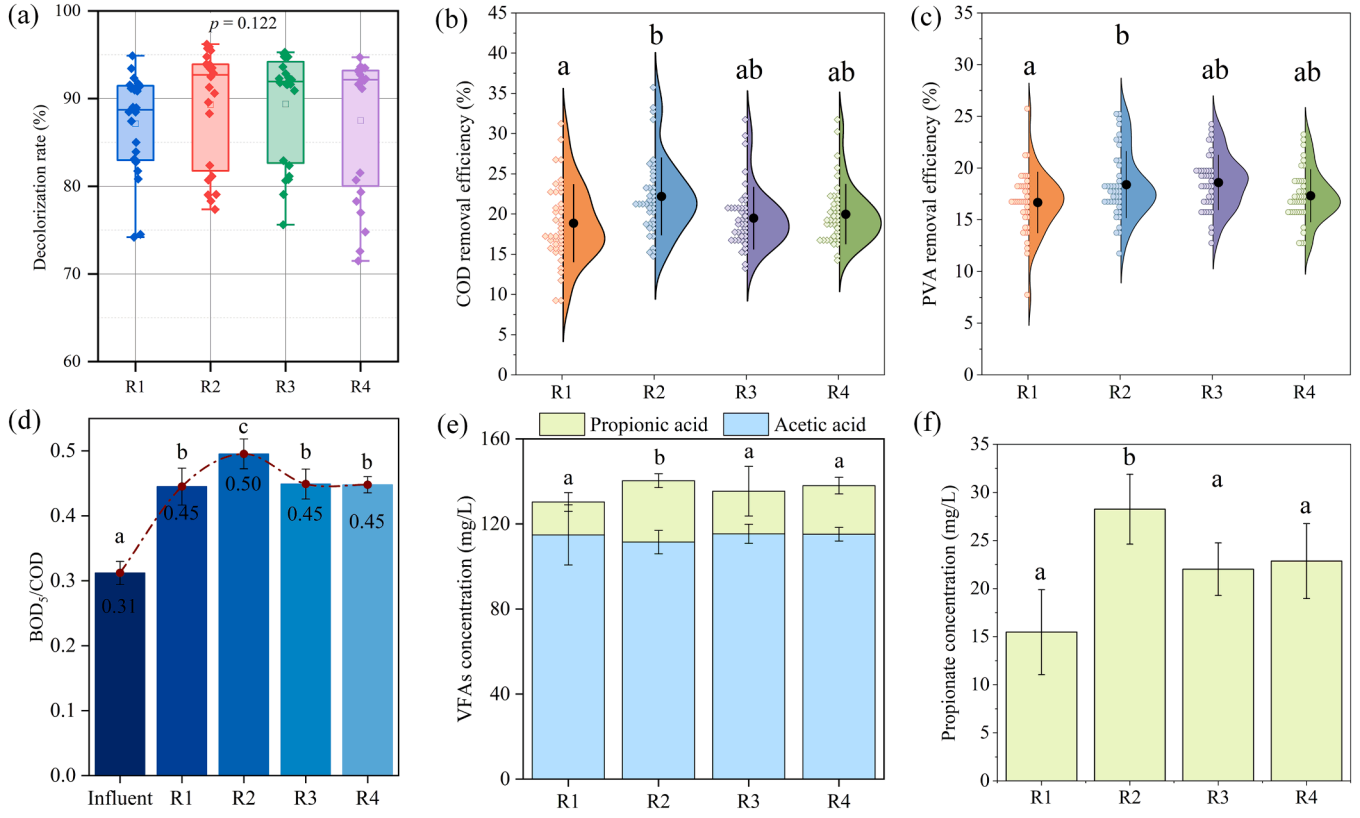


Fig. 3. Degradation and acidogenic performance under different hydrodynamic conditions: (a) decolorization efficiency; (b) COD removal; (c) PVA removal; (d) BOD_5/COD; (e) VFA production; and (f) acidogenesis efficiency. Different lowercase letters indicate significant differences among groups ($p < 0.05$), whereas groups sharing the same letter are not significantly different. The overall difference in decolorization efficiency among R1-R4 was not significant ($p = 0.122$).

more biodegradable forms while maintaining high decolorization efficiency, with R2 showing a particularly favorable overall transformation pattern.

VFA production further supported the observed improvement in wastewater biodegradability (Fig. 3e). Acetate was the dominant VFA, accounting for 79.4%-88.1% of total VFAs, followed by propionate at 11.8%-20.6%. Total VFA production differed significantly among the four operating conditions, with R2 showing the highest average concentration of 140.38 mg/L and significantly higher VFA production than R1, R3, and R4 ($p < 0.05$). Meanwhile, propionate concentration also varied significantly among the operating conditions (Fig. 3f), with the highest mean value of 28.88 mg/L observed in R2. VFAs are key intermediates generated during the hydrolysis and acidification of complex organic substrates and can therefore serve as indicators of acidogenic conversion [33]. The enhanced VFA accumulation in R2, together with its higher BOD_5/COD ratio, suggested that the hydrodynamic conditions under R2 were associated with more effective conversion of complex organic matter into soluble, biodegradable acidogenic products. Notably, the highest VFA production did not coincide with the highest bulk DO or the strongest stirring condition, but occurred under the moderate stirring condition of R2. This observation further suggested that excessive stirring and bulk DO did not necessarily enhance acidogenic conversion, highlighting the importance of maintaining an appropriate balance between oxygen availability and hydrodynamic conditions during MAHA.

3.4. EPS content and composition

EPS are not only microbial metabolic products but also key matrices for cell adhesion, biofilm growth and extracellular reactions [34]. In MAHA systems, EPS can adsorb dye-derived organic matter and provide microenvironments for extracellular enzymatic transformation, thereby facilitating the release of smaller soluble products into the supernatant [35]. As shown in Fig. 4a, R2 had the lowest SL-EPS proportion (48%) and the highest TB-EPS proportion (34%) among the reactors, indicating a shift from loosely bound EPS toward a more compact EPS structure. This restructuring may have improved biofilm stability and extracellular enzyme retention [36]. In addition, PN was the dominant component in all EPS layers, accounting for more than 75.8% (Fig. 4b). The PN/PS ratio, which reflected surface hydrophobicity and aggregation stability of sludge flocs or biofilms [37], remained around 3.4 in R1-R3 but decreased slightly in R4. This was likely caused by excessive shear, which mildly disrupted EPS structure and reduced floc stability [38]. These results indicated that R2 developed a more compact and stable EPS matrix, which may have improved extracellular enzyme retention and supported organic matter transformation and acidogenesis.

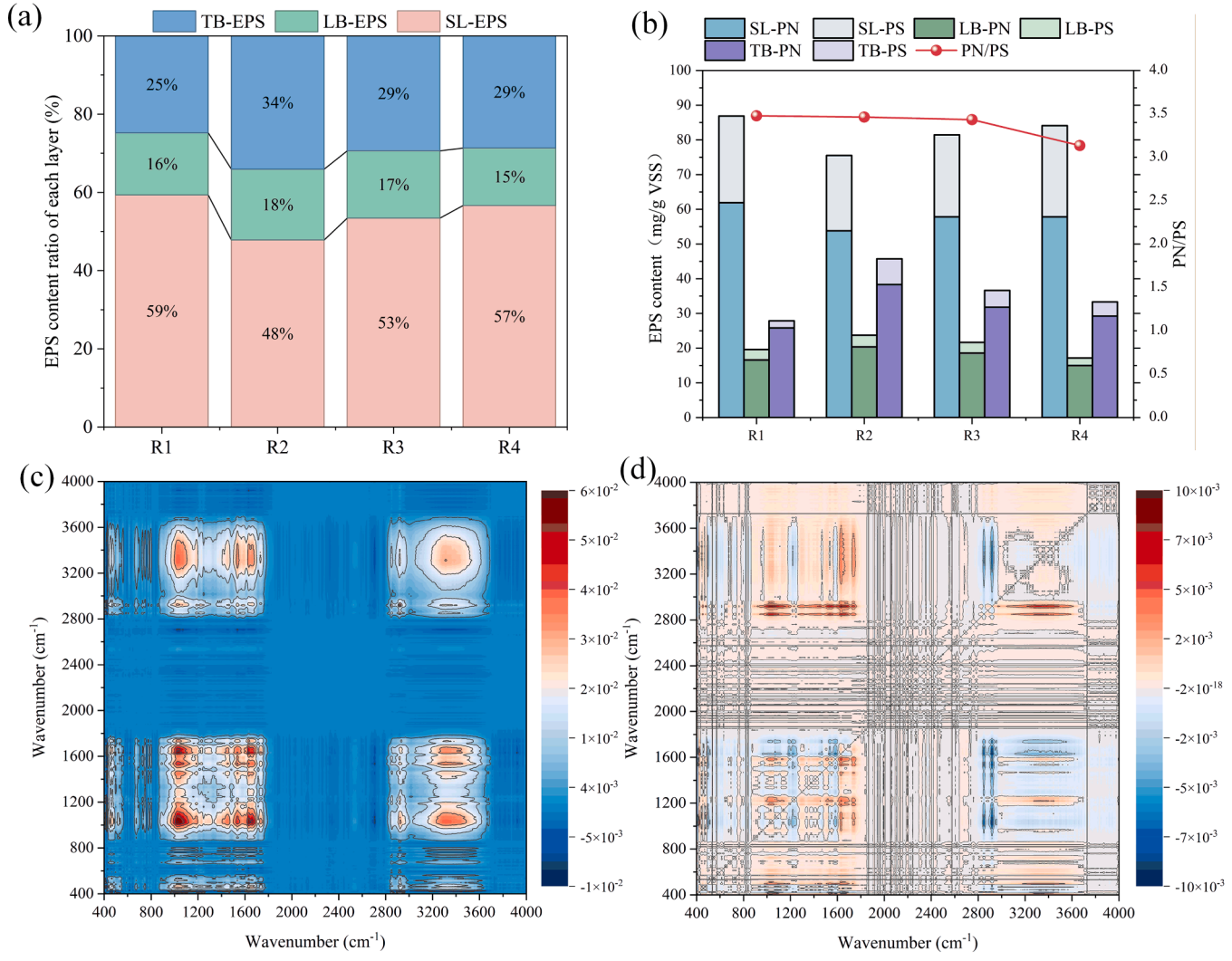


Fig. 4. EPS content and functional group composition. (a) EPS content, (b) Fourier transform infrared spectra, (c) Synchronous 2D correlation spectroscopy, (d) Asynchronous 2D correlation spectroscopy.

3.5. Hydrodynamic regulation of functional group responses

The FTIR spectra of EPS showed similar characteristic peaks within 4000–400 cm^{-1} (Fig. S5), mainly involving O–H, N–H, C–O–C, –N = N–, S═O, benzene ring and substituted benzene structures. The band at 3200–3600 cm^{-1} was assigned to O–H and N–H stretching vibrations [39]. The peaks at 2850–2950 cm^{-1} corresponded to $–CH_3$ and $–CH_2–$ stretching vibrations [40], reflecting aliphatic organic components and protein side chains. The region at 1600–1650 cm^{-1} was associated with aromatic C═C, amide C═O and azo bond (–N = N–) vibrations [40]. The bands at 1400–1550 cm^{-1} were attributed to $–CH_3$ and $–CH_2–$ bending vibrations and aromatic skeleton vibrations. The absorption band near 1360 cm^{-1} was related to O–H bending vibrations in phenolic or alcoholic groups [41]. The 1000–1250 cm^{-1} region represented C–O, C–O–C and S═O stretching vibrations, mainly reflecting EPS polysaccharides and sulfonic groups in dye molecules [42]. The 700–800 cm^{-1} region was assigned to out of plane bending vibrations of benzene rings and substituted benzene structures [43]. The increased intensity of the 700–800 cm^{-1} bands in R2-R4 suggested greater exposure or accumulation of benzene ring and substituted benzene structures during dye degradation.

To further examine the response sequence of functional groups to flow structure in MAHA systems, synchronous (Fig. 4c) and asynchronous (Fig. 4d) 2D-COS spectra were obtained over the range of 4000–400 cm-1. In the synchronous spectrum, autocorrelation was observed for almost all major peaks, with the strongest peaks at 1023, 1624 and 3414 cm^{-1} . The results showed that C–O/C–O–C, –N = N– and O–H/N–H were highly responsive to flow field variations. Cross peaks among the main functional groups were observed in the asynchronous spectrum, indicating that the transformation of these groups in dyeing wastewater did not occur simultaneously. According to Noda's rules [29], the signs of synchronous and asynchronous cross peaks were used to determine the response sequence of the main functional groups (Table S2), which followed the order of $1624 > 1456 > 1360 > 670 = 772 \approx 1186\ cm^{-1}$. The first response at 1624 cm^{-1} indicated that azo bond related conjugated structures were preferentially affected under different turbulence intensities. This was followed by the rearrangement of aliphatic side chains (1456 cm^{-1}) and the bending of phenolic or alcoholic O–H (1360 cm^{-1}). In contrast, the later responses of the bands at 670 and 772 cm^{-1} , together with the 1186 cm^{-1} band, suggested that substituted aromatic structures and S═O containing functional environments were relatively less sensitive to changes in stirring conditions. These observations are consistent with a transformation sequence in which chromophore-associated and conjugated structures respond earlier, whereas changes involving aromatic substituents and sulfur-containing groups occur at later stages. Nevertheless, the 2D-COS results primarily describe the relative spectral response sequence and should not be interpreted as direct evidence for the exact order of individual chemical-bond cleavage or desulfonation reactions.

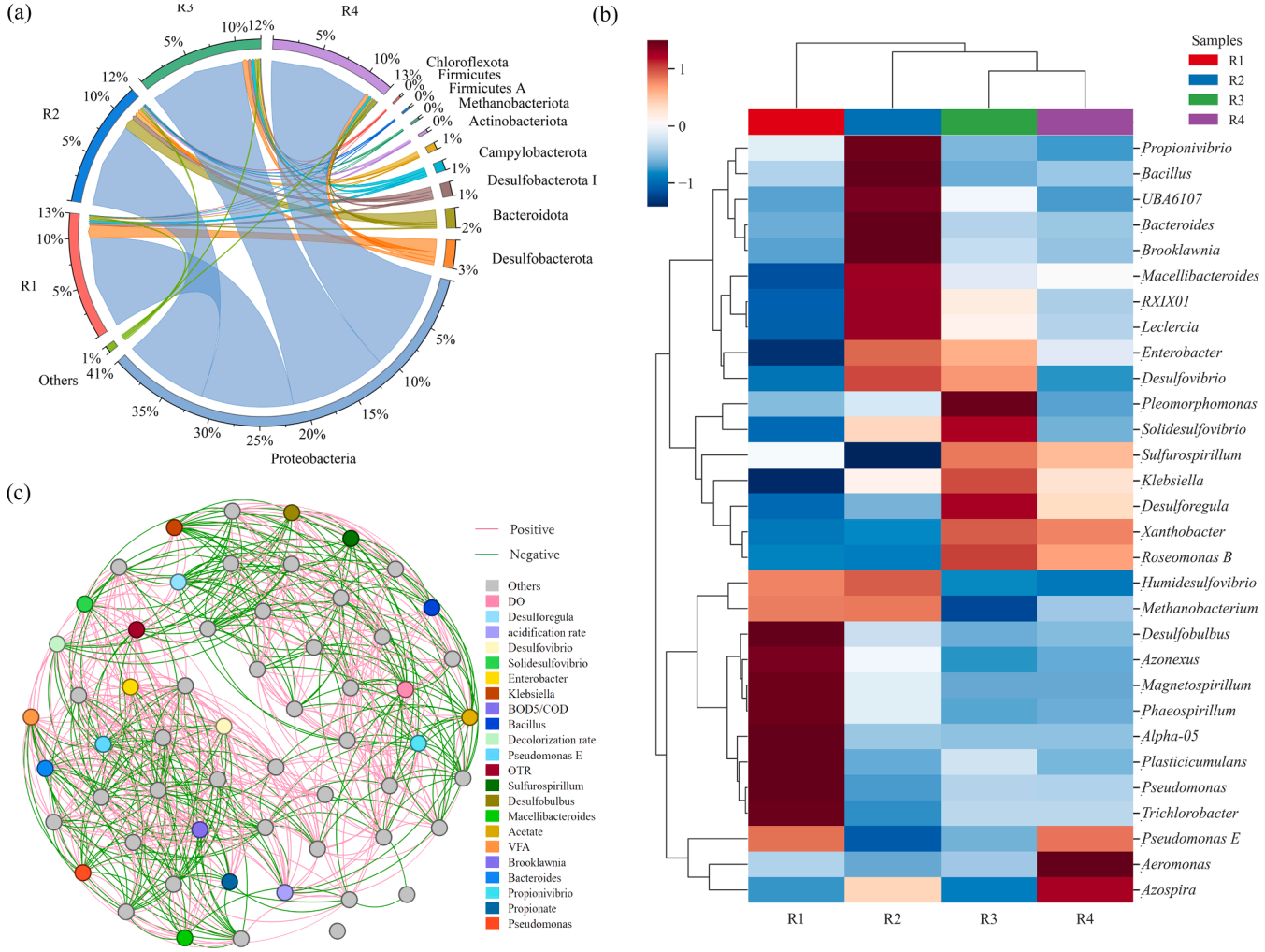


Fig. 5. Microbial community composition and functional taxa associated with hydrolytic acidification. (a) Phylum level, (b) Genus level, (c) Association network between functional genera and process indicators.

3.6. Microbial community composition

Microbial community composition was analyzed at both phylum and genus levels. Proteobacteria was the dominant phylum in all reactors, accounting for 72.91%-87.66% (Fig. 5a). Notably, its relative abundance was the lowest in R2, decreasing by 7.29%, 10.70% and 14.75% compared with R1, R3, and R4, respectively. Conversely, Bacteroidota, Desulfobacterota I, Actinobacteriota, and Firmicutes A were markedly enriched in R2, reaching approximately tenfold the levels observed in the other reactors. These phyla have been associated with the conversion of organic compounds into VFAs [4]. The pronounced shift in community structure therefore suggested that the R2 hydrodynamic condition selectively favored microbial populations potentially involved in organic matter conversion rather than simply increasing the overall abundance of a single dominant phylum.

At the genus level, R2 was enriched in several facultative, hydrolytic, and acidogenic genera (Fig. 5b), including *Klebsiella* (35.79%), *Bacteroides* (7.77%), *Propionivibrio* (6.62%), and *Enterobacter* (5.20%). *Klebsiella* and *Enterobacter* were associated with azo reduction and VFA production, whereas *Bacteroides* and *Propionivibrio* contributed to macromolecular organic matter hydrolysis and propionate fermentation, respectively [44–47]. *Bacillus*, *Brooklawnia*, and *Macellibacteroides* were also enriched in R2. The association network linked these functional genera with oxygen transfer and treatment indicators (Fig. 5c), supporting their potential roles in dye transformation and acidogenesis.

These community shifts corresponded with the oxygen transfer and treatment responses. R1 had the lowest DO and OTR and relatively high abundances of *Plasticicumulans* (9.72%), *Pseudomonas* (7.88%), *Klebsiella* (6.52%), and *Desulfobulbus* (6.10%). In R2, the low DO (0.14 ± 0.02 mg/L) combined with a relatively high OTR (1.37 mg/(L·h)) coincided with the enrichment of facultative, hydrolytic, and acidogenic bacteria and higher VFA production. Although OTR remained at a similar level in R3, the increased DO was accompanied by decreased abundances of *Bacteroides*, *Enterobacter*, *Propionivibrio*, and *Bacillus*. These genera also remained less abundant in R4, which exhibited the highest DO but a lower OTR. Meanwhile, decolorization remained consistently high across R1-R4. These results indicated that R2 provided favorable micro-aerobic conditions for the coexistence of hydrolytic and

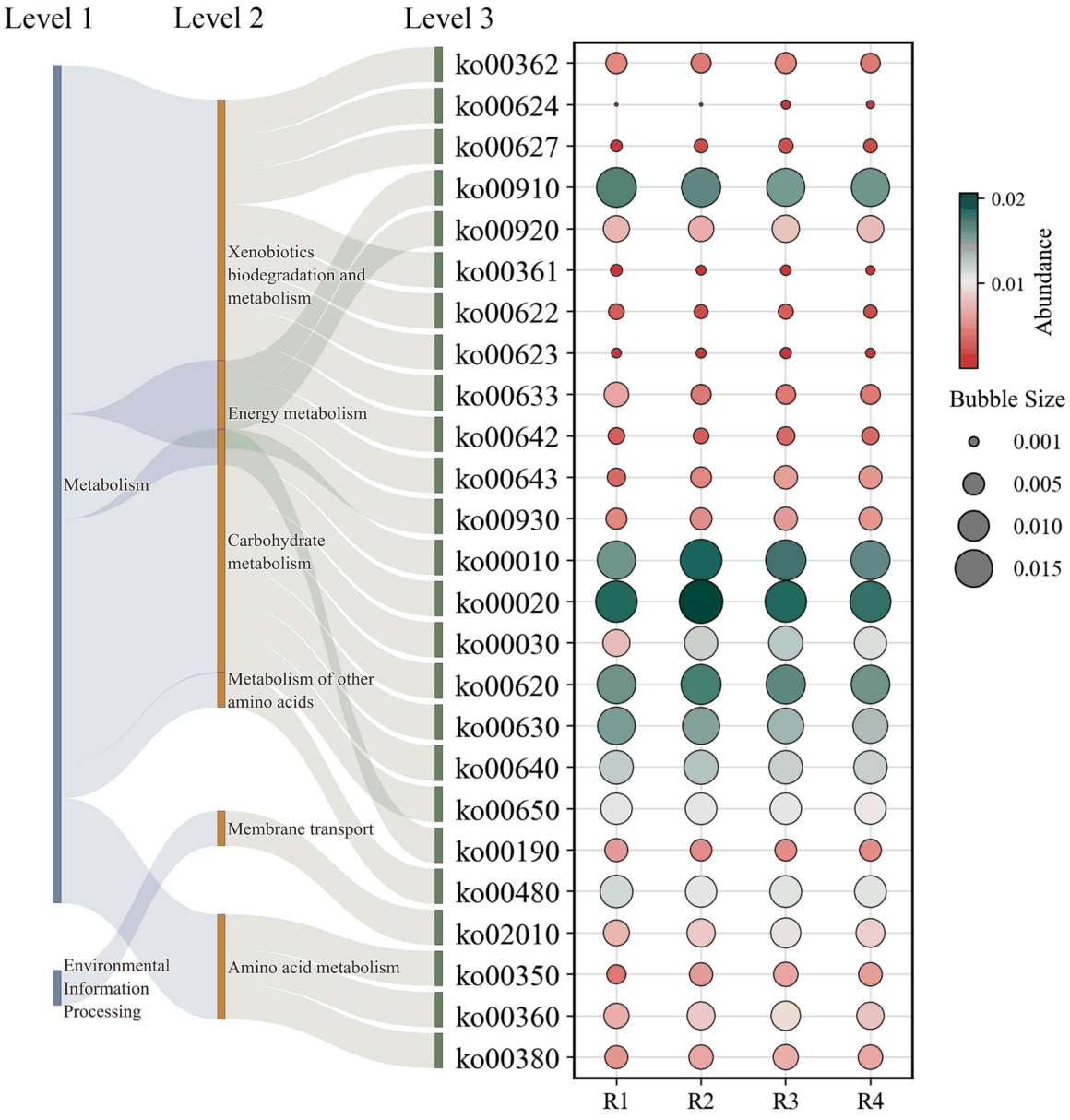


Fig. 6. Functional pathways at KEGG Level 3 involved in Acid Red degradation.

acidogenic populations, supporting VFA production while maintaining effective dye transformation.

3.7. Metabolic pathways

KEGG functional annotation of the metagenomic data was performed to identify key metabolic pathways. Metabolism dominated the annotated functions at level 1, while carbohydrate metabolism was the most abundant pathway at level 2 and peaked in R2 (Fig. S6). KEGG level 3 analysis identified pathways associated with Acid Red degradation and acidogenesis (Fig. 6). Nitrogen metabolism (ko00910) was highly abundant across all reactors, suggesting a potential link with azo bond cleavage and the high decolorization efficiency shown in Fig. 3a. Aromatic compound degradation pathways, including benzoate degradation (ko00362), aminobenzoate degradation (ko00627), and polycyclic aromatic hydrocarbon degradation (ko00624), suggested potential involvement in the transformation of aromatic intermediates following azo reduction. These pathways were further connected with pyruvate metabolism (ko00620) and VFA production related metabolism, such as propanoate metabolism (ko00640), both of which were relatively enriched in R2 and consistent with the increased propionate production (Fig. 3f). Meanwhile, glycolysis/gluconeogenesis (ko00010), the TCA cycle (ko00020), and amino acid metabolism (ko00350, ko00360 and ko00380) may regulate the supply and consumption of carbon intermediates during these processes, thereby maintaining metabolic balance [48]. The enrichment of sulfur metabolism (ko00920) further suggested the potential transformation of sulfonic groups derived from Acid Red. These results suggested that Acid Red transformation was associated with nitrogen and sulfur metabolism, aromatic compound degradation, pyruvate metabolism, acidogenic metabolism, central carbon metabolism, and amino acid metabolism. However, the specific degradation route and catalytic steps require further resolution through key enzyme-encoding genes analysis.

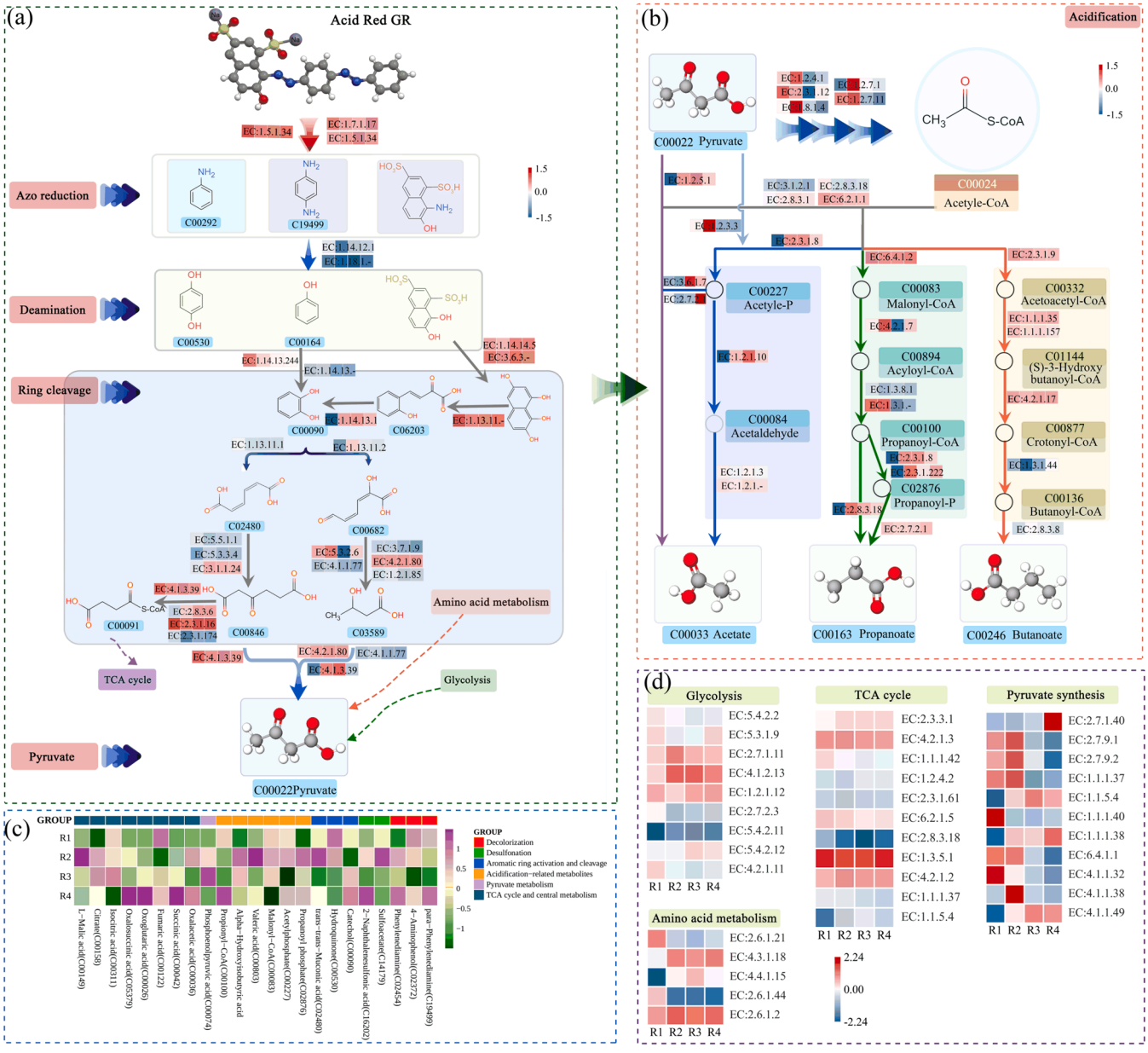


Fig. 7. Key enzyme-encoding genes and metabolites during the degradation of Acid Red GR. (a) Dye degradation pathway, (b) Acid production process, (c) Metabolites; (d) Enzyme-encoding genes related to glycolysis, TCA cycle, pyruvate synthesis, and amino acid metabolism.

3.8. Microbial metabolic functions and dye degradation pathways

Based on the functional group response sequence (Section 3.5), key enzyme-encoding genes, and metabolites identified by LC-MS, a potential pathway for Acid Red GR degradation and acidogenesis in the MAHA system was proposed (Figs. 7a-7b). Acid Red GR transformation was proposed to involve azo bond cleavage, aromatic intermediate transformation, desulfonation, ring cleavage, pyruvate generation and VFA formation.

As shown in Fig. 7a, genes encoding FMN-dependent NADH-azoreductase (EC:1.7.1.17) were relatively abundant across all reactors, suggesting functional potential for reductive azo-bond cleavage. Although oxygen can inhibit azoreductase activity by competing for reducing equivalents [49], microscale DO profiles revealed anaerobic microzones within the carrier-associated biofilms (Fig. S7), providing favorable local conditions for azo reduction despite bulk DO of 0.11–0.24 mg/L. Genes annotated as nitroreductase/dihydropteridine reductase (EC:1.5.1.34) further supported the reductive potential of the microbial communities, consistent with the high decolorization efficiencies (Fig. 3a). Genes encoding anthranilate 1,2-dioxygenase (EC:1.14.12.1 and EC:1.18.1.-) and phenol hydroxylase (EC:1.14.13.244 and EC:1.14.13.-) suggested potential for further transformation of aromatic amine and phenolic intermediates toward catechol (C00090). Hydroquinone (C00530) was also detected by LC-MS (Fig. 7c), although its downstream transformation was less clearly supported by enzyme annotations.

Based on pathway annotation, catechol could undergo ortho cleavage via catechol 1,2-dioxygenase (EC:1.13.11.1) to form cis,cis-muconate (C02480), followed by conversion to 3-oxoadipate (C00846) and succinyl-CoA (C00091). Alternatively, meta cleavage via catechol 2,3-dioxygenase (EC:1.13.11.2) could yield 2-hydroxymuconate semialdehyde (C00682) and subsequently enter pyruvate metabolism. Genes associated with meta cleavage and pyruvate formation were more abundant in R2, suggesting greater potential for channeling aromatic intermediates toward acidogenic metabolism. For sulfonated aromatic intermediates, LC-MS identified 1-naphthol-5-sulfonic acid and the structurally related non-sulfonated intermediate 1-naphthol (C11714), together with intermediates associated with subsequent aromatic transformation and ring cleavage (Table S3). Notably, no ring-cleavage products retaining sulfonate groups were detected. These findings suggested that sulfonated aromatic intermediates underwent desulfonation before ring cleavage and subsequently entered catechol metabolism.

Pyruvate is a key precursor for acidogenic fermentation [50] and can be converted to acetyl-CoA and subsequently to acetate through several enzymatic steps (Fig. 7b). The relatively high abundance of acetate-forming enzyme genes was consistent with the dominance of acetate among VFAs. In R2, enrichment of genes encoding propionate-related enzymes (EC:2.3.1.8 and EC:2.8.3.18) and intermediates, including malonyl-CoA and propanoyl-CoA, coincided with the highest propionate production (Fig. 3f), suggesting enhanced propionate fermentation. In contrast, the low abundance of downstream butyrate-pathway enzyme genes was consistent with limited butyrate formation.

Glycolysis/gluconeogenesis, the TCA cycle, and amino acid metabolism were interconnected through central metabolites such as pyruvate and acetyl-CoA, linking aromatic degradation with acidogenic metabolism. Genes associated with the TCA cycle and amino acid metabolism showed only minor differences among reactors (Fig. 7d), whereas several glycolytic and pyruvate-forming enzyme genes were relatively enriched in R2. These patterns suggested a greater metabolic potential for channeling carbon toward pyruvate in R2, thereby supporting the connection between aromatic intermediate transformation and VFA formation.

3.9. Integrated association analysis

To further explore the integrated association, genes annotated as encoding key enzymes potentially involved in Acid Red GR transformation and acidogenesis, and their top ten contributing genera were identified (Fig. S8). A microbe-enzyme-metabolite correlation network was then constructed (Fig. 8a). Cross-group correlations (150) were markedly more frequent than within-group correlations (83), and positive correlations predominated (61.83%). This suggested coordinated associations among functional microorganisms, enzyme-encoding genes, and metabolite profiles during Acid Red GR transformation and VFA formation.

Bacteroides, *Klebsiella*, *Propionivibrio*, *Bacillus*, and *Pseudomonas* were identified as key genera linking enzyme modules with metabolite variations. Among them, *Pseudomonas* was strongly correlated with aromatic transformation-related enzymes ($r > 0.783$) and intermediates such as hydroquinone ($r > 0.817$), suggesting potential involvement in aromatic intermediate transformation. In contrast, *Bacteroides*, *Propionivibrio*, and *Bacillus* were closely associated with enzymes related to acidogenesis and central carbon metabolism ($r > 0.794$), supporting their potential roles in linking aromatic degradation products with VFA formation. Notably, *Klebsiella*-associated enzyme modules showed contrasting correlations with 4-aminophenol and hydroquinone, suggesting a complex relationship between *Klebsiella* and aromatic intermediate transformation.

Mantel tests were used to assess relationships between environmental factors and microbial metabolic responses (Fig. 8b). Turbulence intensity was significantly associated with functional microorganisms and key enzyme-encoding genes, which were also positively correlated with oxygen transfer parameters (k_La, SOTR, SOTE, and OTR). The results indicated that the flow regime was associated with shifts in microbial composition and enzyme-encoding gene profiles related to hydrolysis and acidogenesis. However, flow parameters were not significantly correlated with oxygen transfer, indicating that oxygen transfer characteristics were not determined by turbulence intensity alone. R1 exhibited a relatively uniform but weak flow field, whereas R3 and R4 showed stronger but more heterogeneous turbulence. R2 combined relatively uniform flow with moderate intensity, consistent with its favorable oxygen-transfer and acidogenic responses. Metabolites showed comparatively weak correlations with oxygen-transfer parameters and VFAs, suggesting that their profiles reflected multiple processes of formation, transformation, and consumption rather than oxygen transfer alone.

3.10. Potential mechanisms of hydrodynamic regulation in the MAHA system

Based on these results, a potential mechanism linking flow regime, oxygen mass transfer, dye degradation, hydrolysis acidification, and microbial metabolism in the MAHA systems was proposed (Fig. 8c). R2 combined appropriate turbulence intensity with greater flow uniformity and maintained relatively high oxygen transfer performance (k_La and OTR). This hydrodynamic condition was associated with the coexistence of facultative anaerobic and fermentative bacteria, including *Bacteroides*, *Klebsiella*, *Propionivibrio*, *Bacillus*, *Enterobacter*, and *Brooklawnia*. The enrichment of these functional taxa was accompanied by higher abundances of genes encoding enzymes related to azo-bond reduction, aromatic intermediate transformation, and acidogenic metabolism, thereby promoting Acid Red GR decolorization, aromatic intermediate conversion, biodegradability improvement, and VFA production.

Although the synthetic wastewater was designed to reflect real dyeing wastewater, it cannot fully capture its compositional complexity and temporal variability. Nevertheless, the controlled matrix enabled reliable elucidation of the relationships among hydrodynamics, oxygen transfer, microbial metabolism, and Acid Red GR transformation. Future studies should validate these mechanisms using real wastewater under

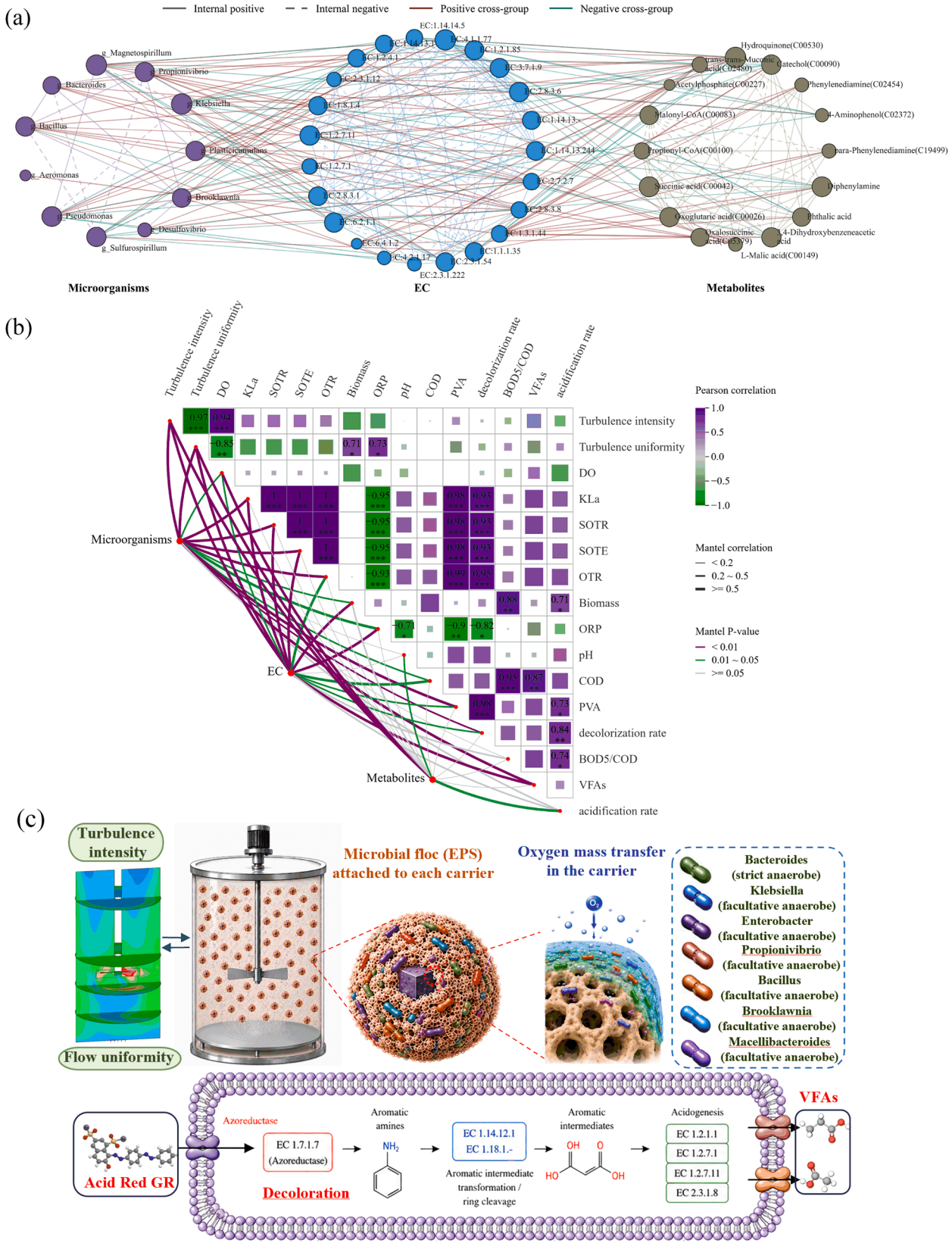


Fig. 8. The potential metabolic mechanisms of microorganisms in the MAHA reactor. (a) The network correlation between microorganisms and key enzymes and metabolites, (b) Mantel test analysis, (c) Potential mechanism by which flow conditions regulate oxygen mass transfer and microbial metabolism.

long-term variable conditions and further verify the proposed metabolic pathways through proteomics and enzyme activity assays.

4. Conclusion

This study demonstrated that hydrodynamic regulation was an important factor controlling oxygen transfer, microbial adaptation, and Acid Red GR transformation in the MAHA reactor. Moderate stirring intensity (R2, 120 rad/min) established a favorable low-oxygen microenvironment by balancing turbulence enhancement and flow uniformity, thereby improving biodegradability and acidogenic conversion while maintaining efficient decolorization. Multi-scale analyses indicated that this condition promoted the formation of a stable EPS matrix, enriched facultative hydrolytic and acidogenic microorganisms, and enhanced functional potentials associated with azo-dye transformation and carbon metabolism. Integrating LC-MS, FTIR/2D-COS, and microbial functional information, a potential Acid Red GR transformation pathway involving azo-bond reduction, aromatic structure transformation, and subsequent acidogenesis was proposed. These findings highlight that optimizing hydrodynamic conditions, rather than simply increasing oxygen supply, is a promising strategy for regulating microbial metabolism and enhancing MAHA performance in dyeing wastewater treatment.

Environmental implication

Acid Red GR is a hazardous azo dye because its chromophoric azo bonds, sulfonated aromatic structures, and aromatic amine-derived intermediates make dyeing wastewater persistent, colored, and potentially toxic. This work shows that hydrodynamic regulation in a microaerobic hydrolysis acidification reactor can steer oxygen transfer, enrich facultative fermentative bacteria, and activate enzymes for azo reduction, desulfonation, aromatic ring cleavage, and acidogenesis. By converting refractory dye-derived organics into more biodegradable products and VFAs under moderate stirring, the study provides an operational strategy for improving industrial dyeing wastewater pretreatment and reducing environmental risks from hazardous organic pollutants.

CRediT authorship contribution statement

Jiayao Hui: Writing – original draft, Methodology, Funding acquisition. **Min Wang:** Investigation. **Wen Cheng:** Supervision, Methodology. **Tian Wan:** Methodology. **Ziyi Jia:** Methodology, Data curation. **Yong Zhang:** Methodology. **Jiehui Ren:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Xiaoxin Yang:** Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2026.143745.

Data availability

Data will be made available on request.

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